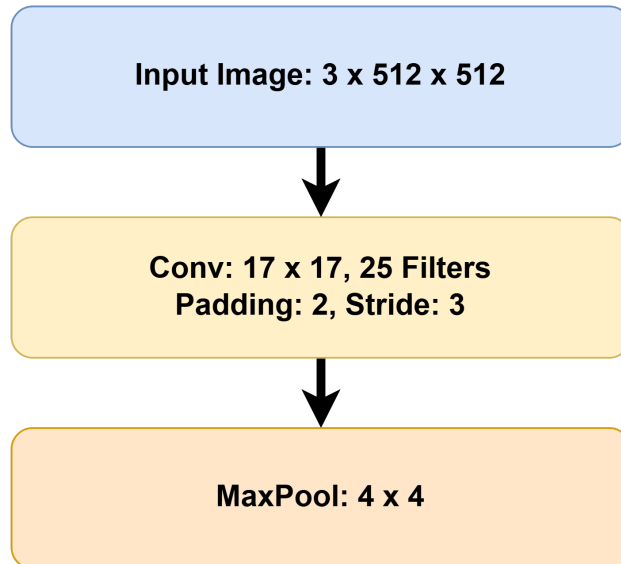


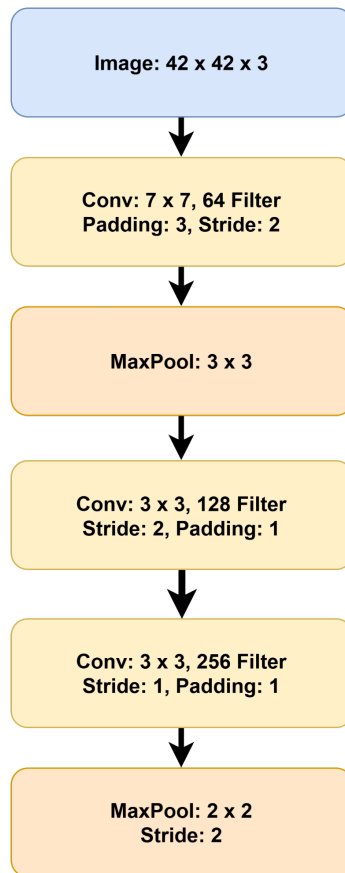
1. Input Image Size: 256x256x3 is passed through a Convolution Layer with 15 filters of size 9x9. Find the output size after the convolution layer. Find the total number of parameters.

2. Below you can find CNN to classify the image between SUV and SEDAN car.



Find the output size after the MaxPool.

3. For the Network below find the Output size of the last layer. Find the total parameters.



4. For the below image show what is the result of convolution:

IMAGE			
1	1	0	1
1	0	0	1
0	1	1	1
1	1	1	0

KERNEL	
3	1
0	1

5. From the result of question number 4 implementing Standard Batch Normalization, the formula is given below.

$$x_{normalized} = \frac{x_{current} - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

$\mu = mean$

$\sigma^2 = variance$

$\epsilon = 0.00001$

Show the final output.

6. For the image in question 4 apply 2 x 2 MaxPooling and show the final result.